15. Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame.

Aim

To filter and display the rows containing at least two missing values using Pandas.

Procedure

1. Import Pandas and NumPy.
2. Create a DataFrame with missing values.
3. Count the NaN values in each row.
4. Select rows having two or more NaN values.
5. Display the result.

Program and output

```
import pandas as pd
import numpy as np
# Create DataFrame
data = {
    "A": [10, np.nan, 30, np.nan],
    "B": [20, np.nan, np.nan, 40],
    "C": [np.nan, 50, np.nan, 60],
    "D": [40, 70, 80, np.nan]
}
df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)
# Select rows with at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]
print("\nRows with at least 2 NaN values:")
print(result)
```

```
Original DataFrame:
      A     B     C     D
0  10.0  20.0   NaN  40.0
1   NaN   NaN  50.0  70.0
2  30.0   NaN   NaN  80.0
3   NaN  40.0  60.0   NaN

Rows with at least 2 NaN values:
      A     B     C     D
1   NaN   NaN  50.0  70.0
2  30.0   NaN   NaN  80.0
3   NaN  40.0  60.0   NaN
```

Result

Thus, the rows containing at least two NaN values were displayed successfully.

16. Write a Pandas program to split the dataframe into groups based on school code. Also check the type of GroupBy object.

Aim

To group the DataFrame based on school code and find the type of GroupBy object using Pandas.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing school details.
3. Group the data using school code.
4. Check the type of GroupBy object.
5. Display the groups.

Program and output

```
import pandas as pd

# Create DataFrame

data = {

    "School_Code": ["S01", "S02", "S01", "S03", "S02"],

    "Student_Name": ["John", "Alice", "David", "Emma", "Robert"],

    "Marks": [85, 90, 75, 88, 92]

}

df = pd.DataFrame(data)

print("Original DataFrame:")
```

```
print(df)

# Group by School Code

group = df.groupby("School_Code")

# Check type

print("\nType of GroupBy object:")

print(type(group))

# Display groups

print("\nGroups:")

for name, data in group:

    print(name)

    print(data)
```

```
Original DataFrame:
  School_Code Student_Name  Marks
0         S01         John     85
1         S02        Alice     90
2         S01        David     75
3         S03         Emma     88
4         S02       Robert     92

Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>

Groups:
S01
  School_Code Student_Name  Marks
0         S01         John     85
2         S01        David     75
S02
  School_Code Student_Name  Marks
1         S02        Alice     90
4         S02       Robert     92
S03
  School_Code Student_Name  Marks
3         S03         Emma     88
```

Result

Thus, the DataFrame was successfully divided into groups based on school code, and the GroupBy object type was identified.